



ASIAN
INSTITUTE OF
MANAGEMENT

Data Mining and Wrangling

Representative-based Clustering

Session 21 and 22

BSDSBA 2028

08 April 2026

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Class Administrative Matters

Course Deliverables

Assessed – ~~Midterm Exam~~, ICA04, ICA05, **ICA06**

Due this Week – MP1 Peer Eval (Fri April 10), ICA09 (Wed April 8), **E7 (Fri April 17)**

Upcoming – R10 (Wed April 8), ICA10 (Wed April 15), E8 (Fri April 17), **A2 (Fri April 24)**

For Checking – E3, R07, E4, E3S2, ICA07, E5, A1, R08, ICA08, E6, MP1, R09



Class Administrative Matters

Final Project

Schedule – May 7 (Thursday) 1:30 PM to 5:00 PM

Requirements (DMW) – Data preparation (collection thru scraping/web API, proper data storage, data wrangling, data preprocessing or feature reduction using PCA/SVD), Code organization, Final Project Report

Topic Proposals – April 17 (Friday) EOD



Session 21 and 22 – Representative-based Clustering

Gameplan

Session 21

Introduction to Clustering

Representative-based Clustering

Session 22

Applied Example of Representative-based Clustering

In-Class Activity 10 – Representative-based Clustering



Session 23 and 24 – Hierarchical Clustering

Gameplan

Session 23

Agglomerative Methods

Divisive Methods

Session 24

Applied Example of Hierarchical Clustering

In-Class Activity 11 – Hierarchical Clustering



Session 25 and 26 – Density-Based Clustering

Gameplan

Session 25

DBSCAN

Evaluation of Clustering Methods

Session 26

Applied Example of DBSCAN

In-Class Activity 12 – Density-based Clustering



Introduction to Clustering



Deja Vu (2019)

Dave Rodgers



超える

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Introduction to Clustering

What is Clustering?

Clustering is the process of partitioning a set of data objects (or observations) into subsets.



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Introduction to Clustering

What are applications of Clustering?

Customer segmentation – Analyze the common behaviors of groups of similar customers.

Web Search – Organizing the search results in a concise and accessible manner.

Image Recognition – Grouping photos of the same person into a group.

Data Preprocessing – Clustering can be used as preprocessing step for classification and outlier detection models.



Introduction to Clustering

What are the different clustering methods?

Representative-based – A set of partitioning *representatives* is used to directly cluster data points based on proximity with the representatives.

Hierarchical Methods – A hierarchical decomposition is created from a given set of data objects.

Density-based and Grid-based – Clusters are continuously formed as long as the density in the “neighborhood” exceeds some threshold.

Introduction to Clustering

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Introduction to Clustering

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“English”



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“Japanese”


“French”



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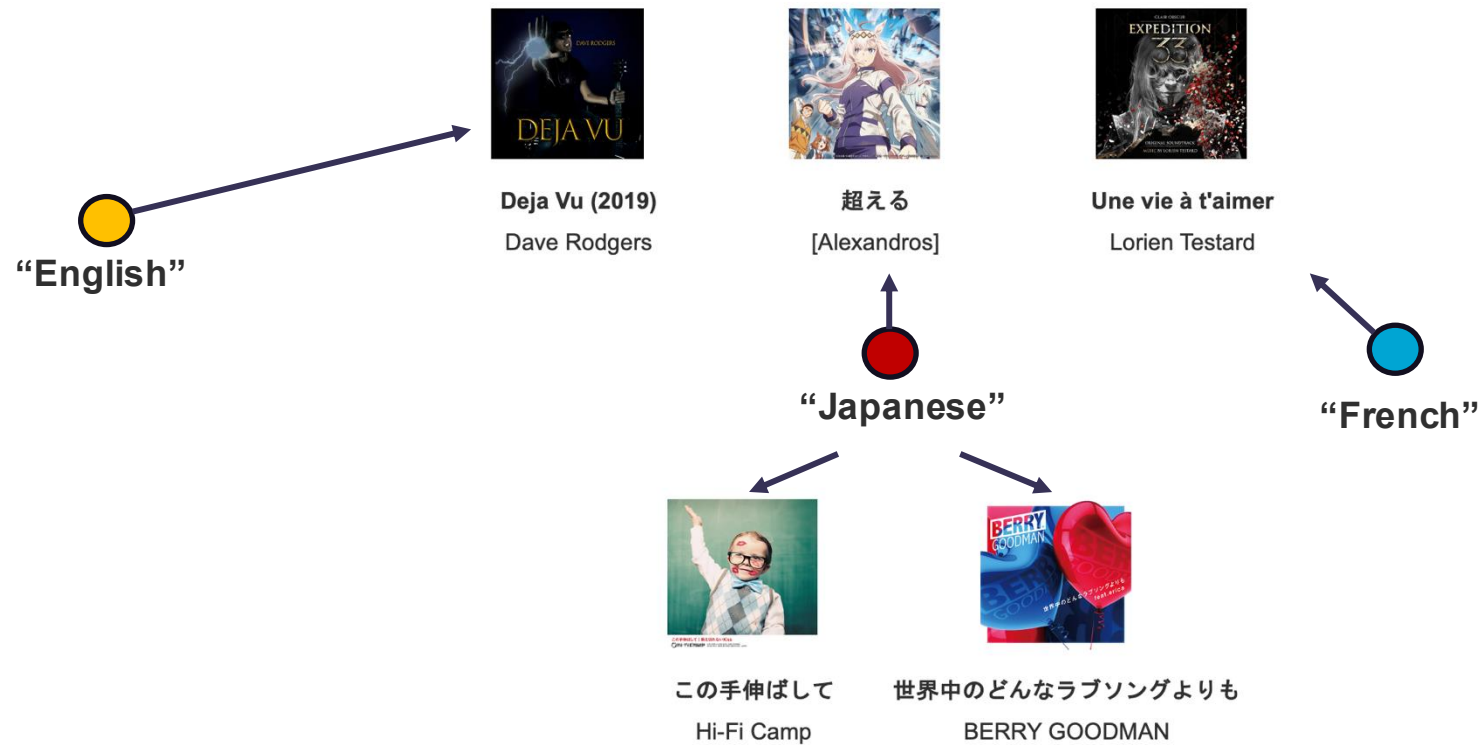


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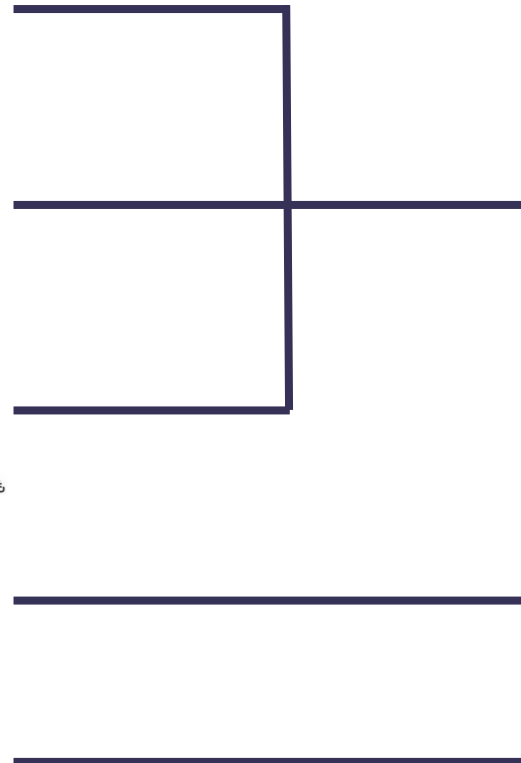
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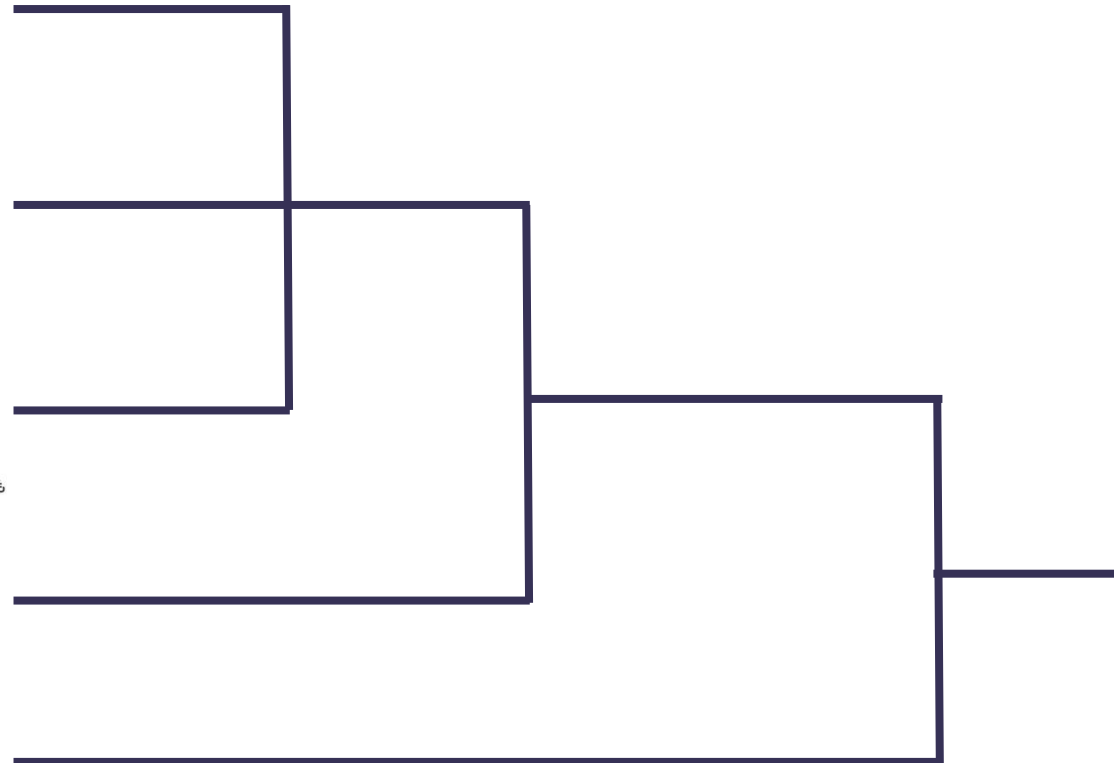
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Introduction to Clustering

What are the different clustering methods?

Density-based Methods – Clusters are continuously formed as long as the density in the “neighborhood” exceeds some threshold.



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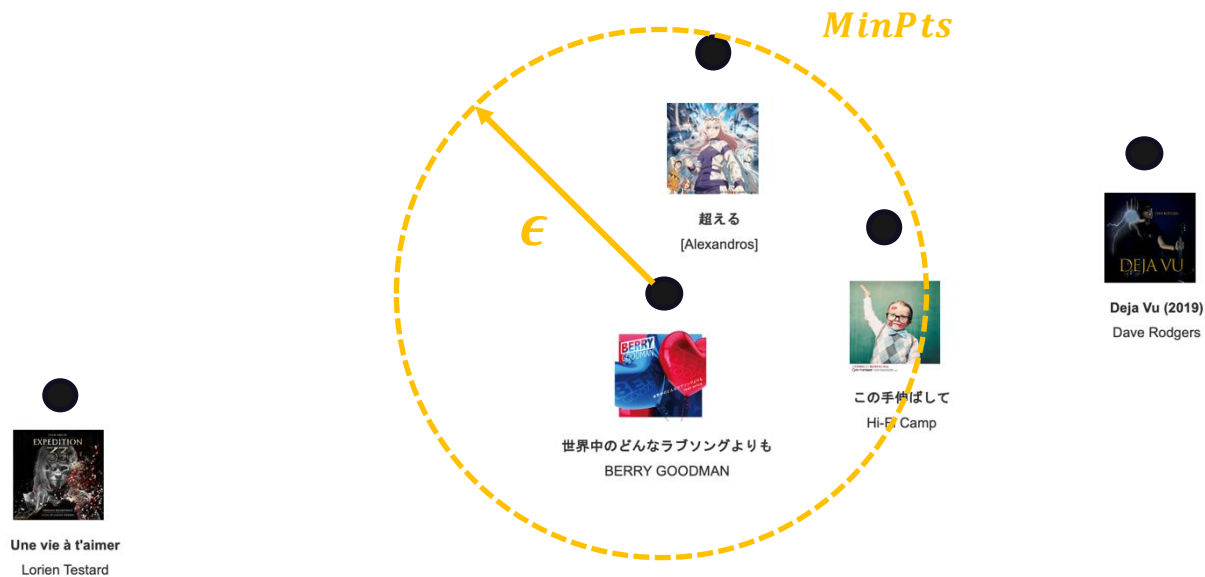
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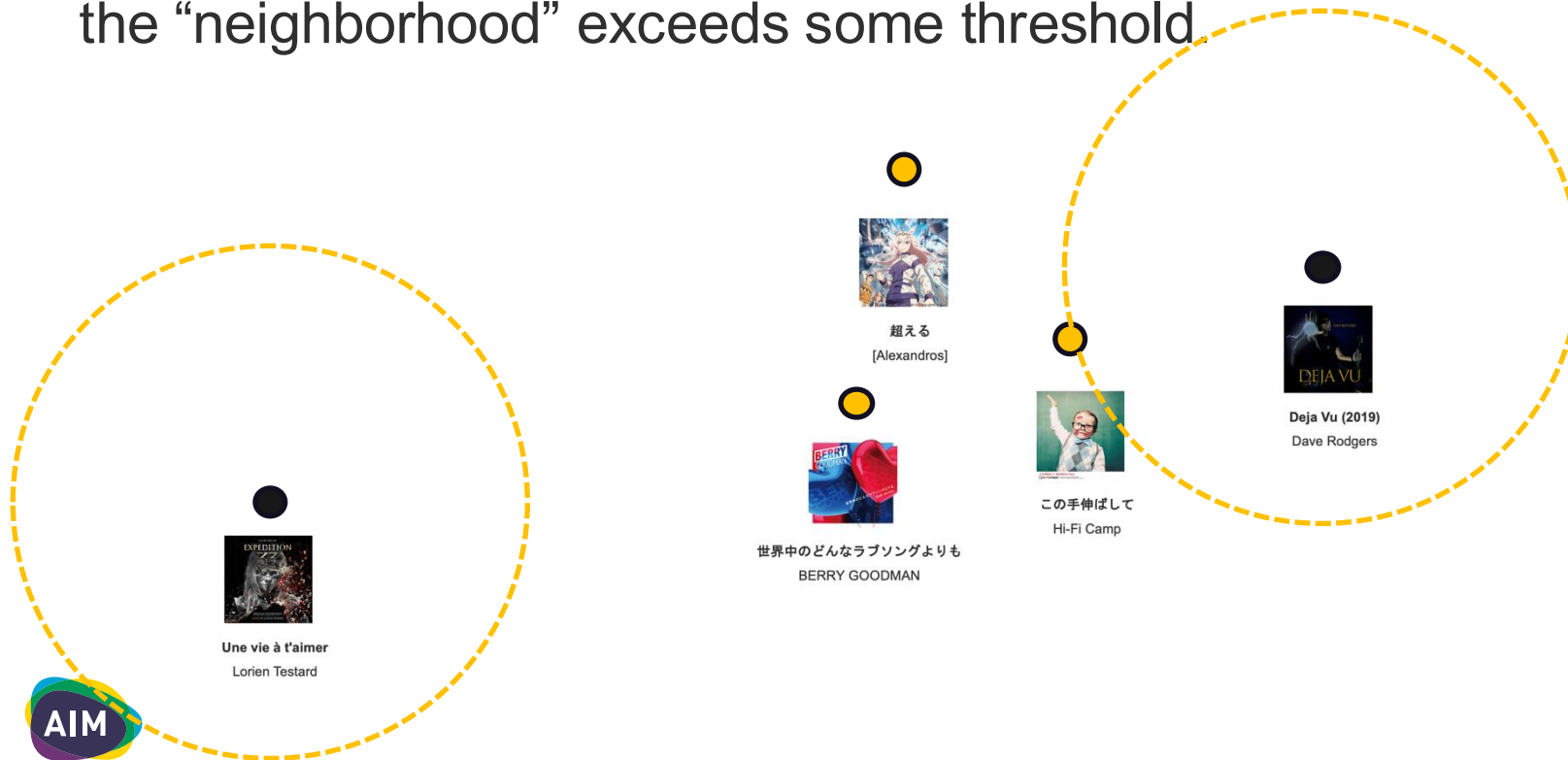
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Introduction to Clustering

Challenges to Keep in Mind

Hyperparameters – Domain knowledge might be needed to inform hyperparameters such as number of clusters.

Outliers and noisy data – Clustering algorithms can be sensitive to noise such as outliers and erroneous data.

Scalability – As dataset size grows, clustering algorithms can be computationally expensive.

Introduction to Clustering

Challenges to Keep in Mind

High-Dimensional Data – Finding clusters in high dimensional data is challenging due to the curse of dimensionality.

Arbitrarily Shaped Clusters – Clustering algorithms based on Euclidean or Manhattan distance will tend to find spherical clusters with similar size or density.

Interpretability and Usability – Clustering need to be anchored on insights that are meaningful for the users.



Introduction to Clustering

Challenges to Keep in Mind

Mixed Data Types – Many applications require analysis on mixed data types.

Incremental Clustering – Incorporating new data can be challenging for some clustering algorithms and may be sensitive to the input data order.

Representative-based Clustering

Representative-based Clustering

Representative-based Clustering Definition

Consider a data set $\mathcal{D}$ containing n data points denoted by $\overline{X}_1, \overline{X}_2, \dots, \overline{X}_n$ in d -dimensional space. The goal is to determine k ($k \leq n$) representatives $\overline{Y}_1, \overline{Y}_2, \dots, \overline{Y}_k$, that minimizes the objective function O :

$$O = \sum_{i=1}^n \left[\min_j Dist(\overline{X}_i, \overline{Y}_j) \right]$$

Representative-based Clustering

Generic k -representatives Algorithm

The generic k -representatives approach starts by initializing the k representatives, then refines the representatives and clustering assignment, iteratively:

- **Assign Step** – Assign each data point to its closest representative using a distance function.
- **Optimize Step** – Determine the optimal representative $\bar{Y}_j$ for each cluster C_j that minimizes its *local* objective function $\sum_{\bar{X}_i \in C_j} \text{Dist}(\bar{X}_i, \bar{Y}_j)$

Representative-based Clustering

The k -Means Algorithm

In the **k -means algorithm**, the sum of the squares of the Euclidean distances of data points to their closest representatives is used to quantify the objective function of the clustering:

$$Dist(\overline{X_i}, \overline{Y_j}) = \|\overline{X_i} - \overline{Y_j}\|_2^2$$

Each representative is assigned to be the *mean* value of the objects in the cluster.

Representative-based Clustering

The k -Means Algorithm

Input: k – number of clusters; D – a data set containing n data points

Output: A set of k clusters

Method: 1. Arbitrarily choose k data points as the initial cluster centers;

2. **Repeat (until convergence)**

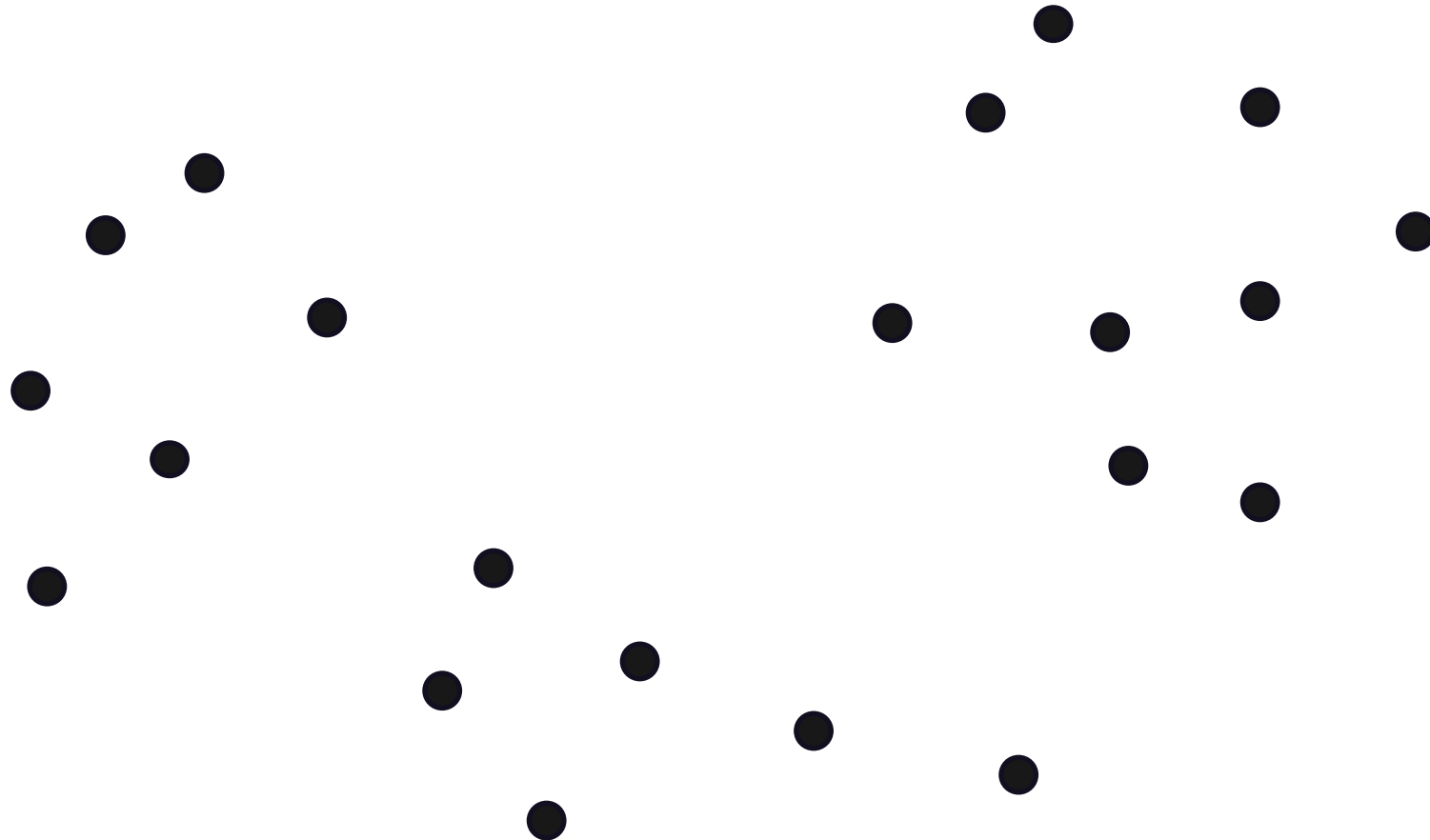
1. (Re)assign each data point to its nearest representative

2. Update cluster centers by calculating the mean value of each data point for each cluster.



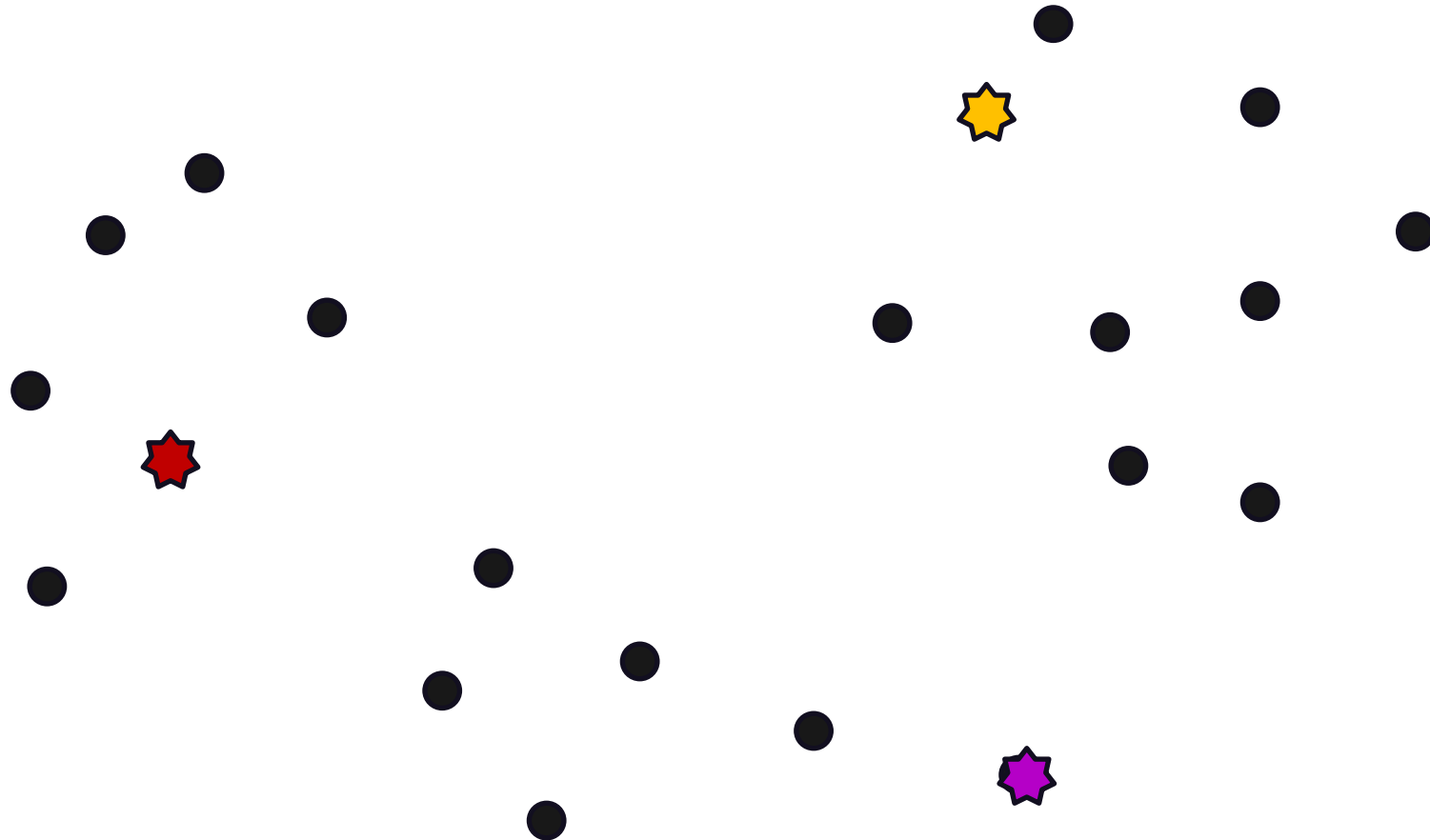
Representative-based Clustering

The k -Means Algorithm



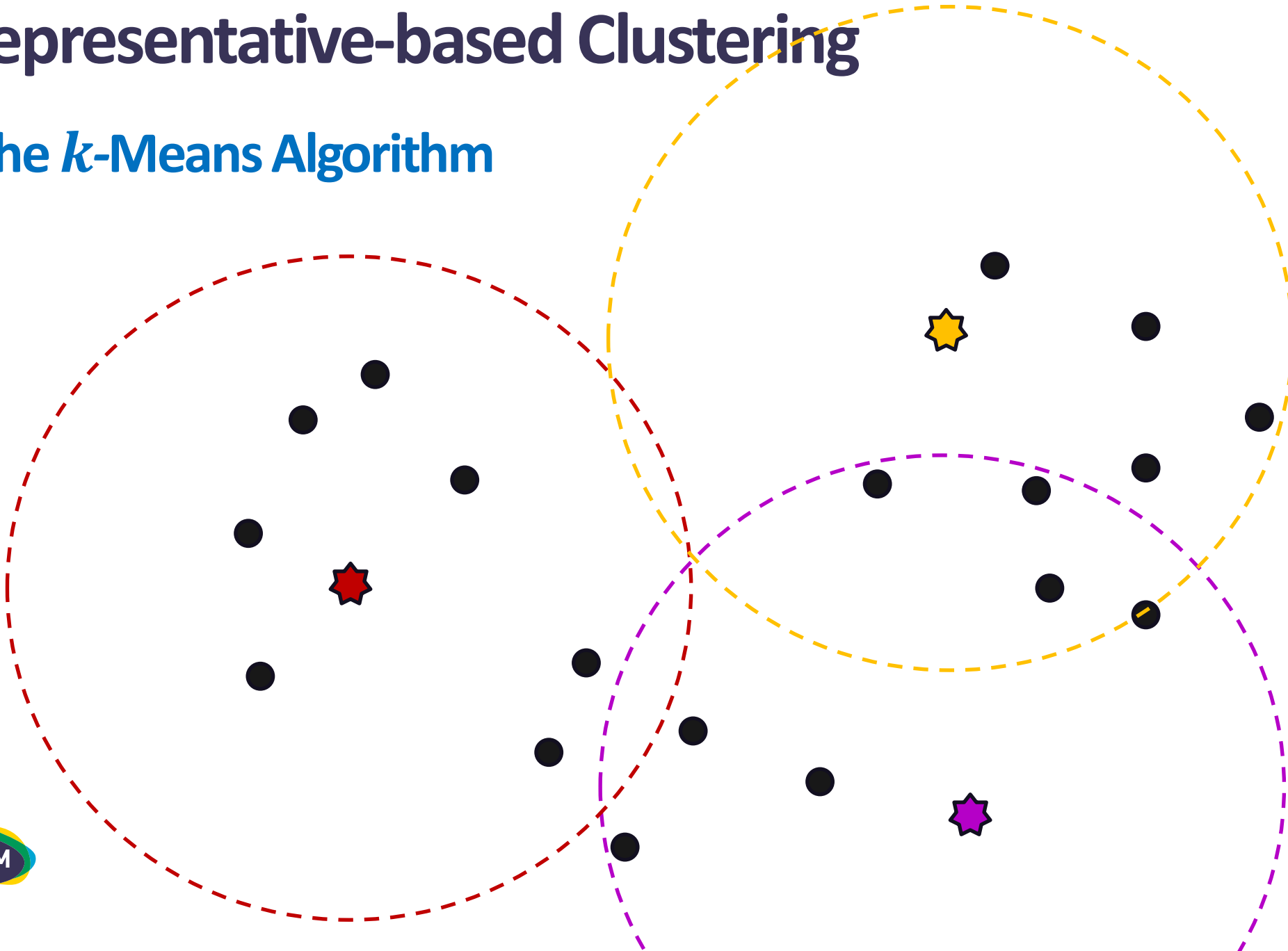
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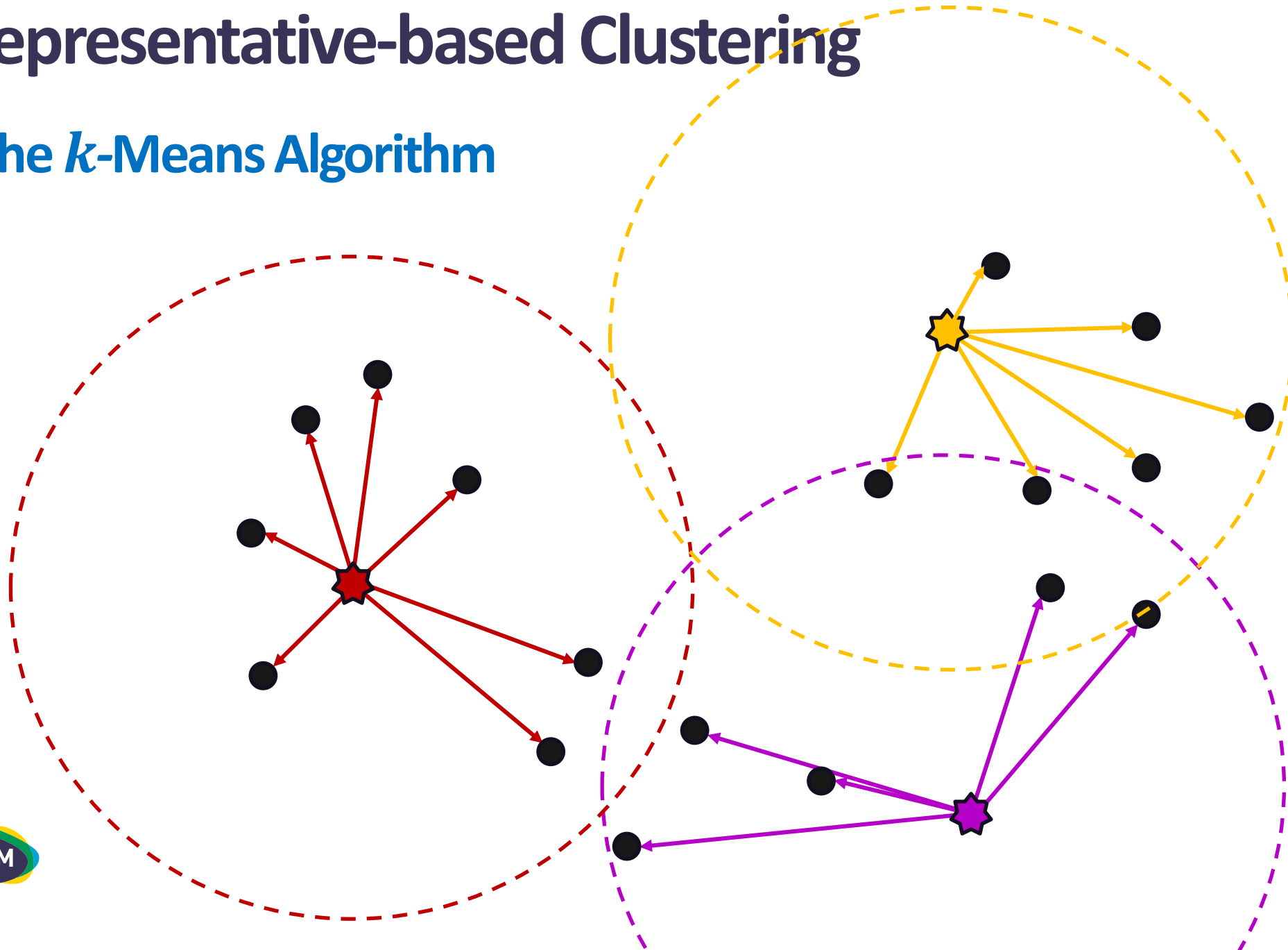
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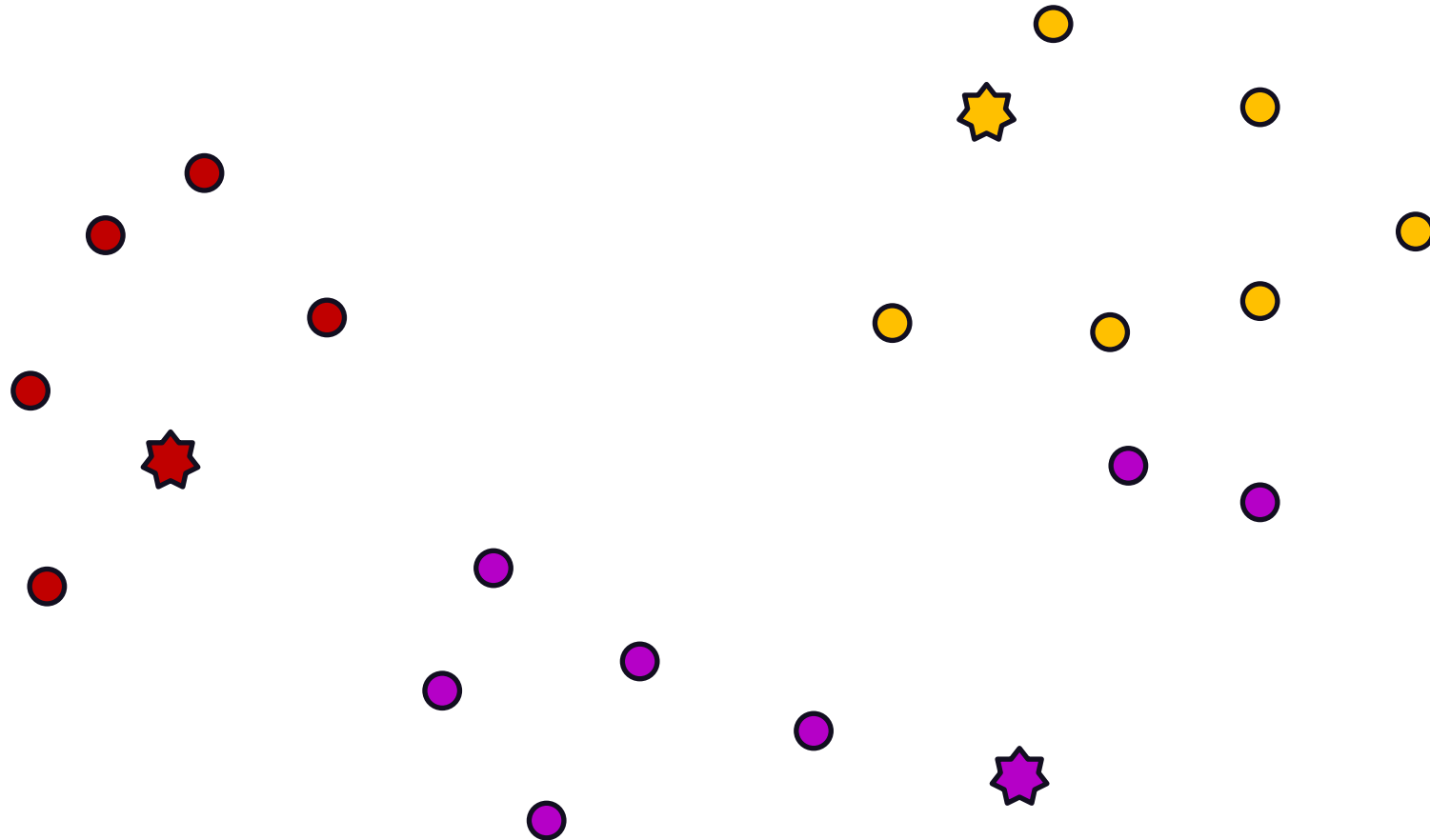
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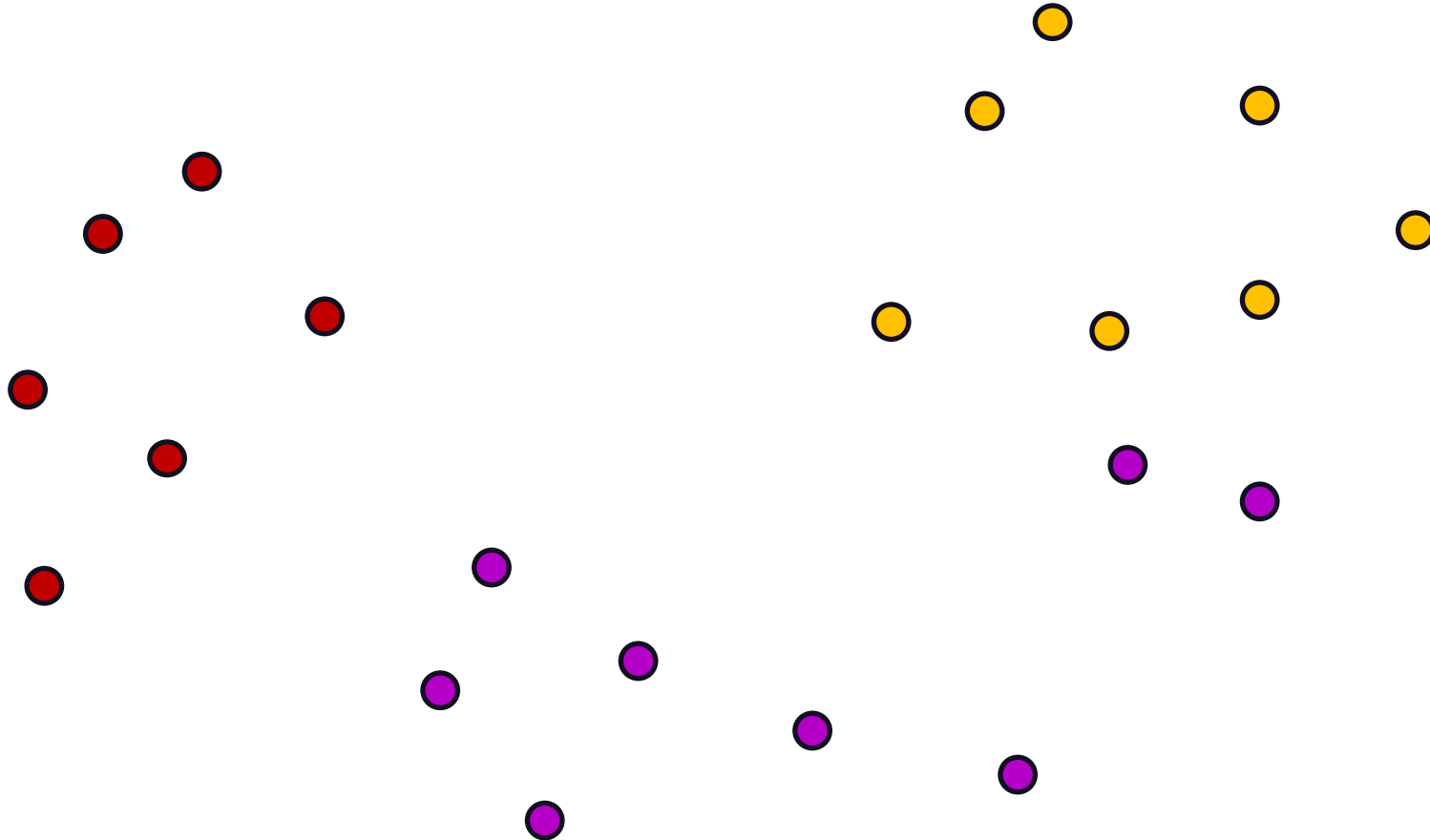
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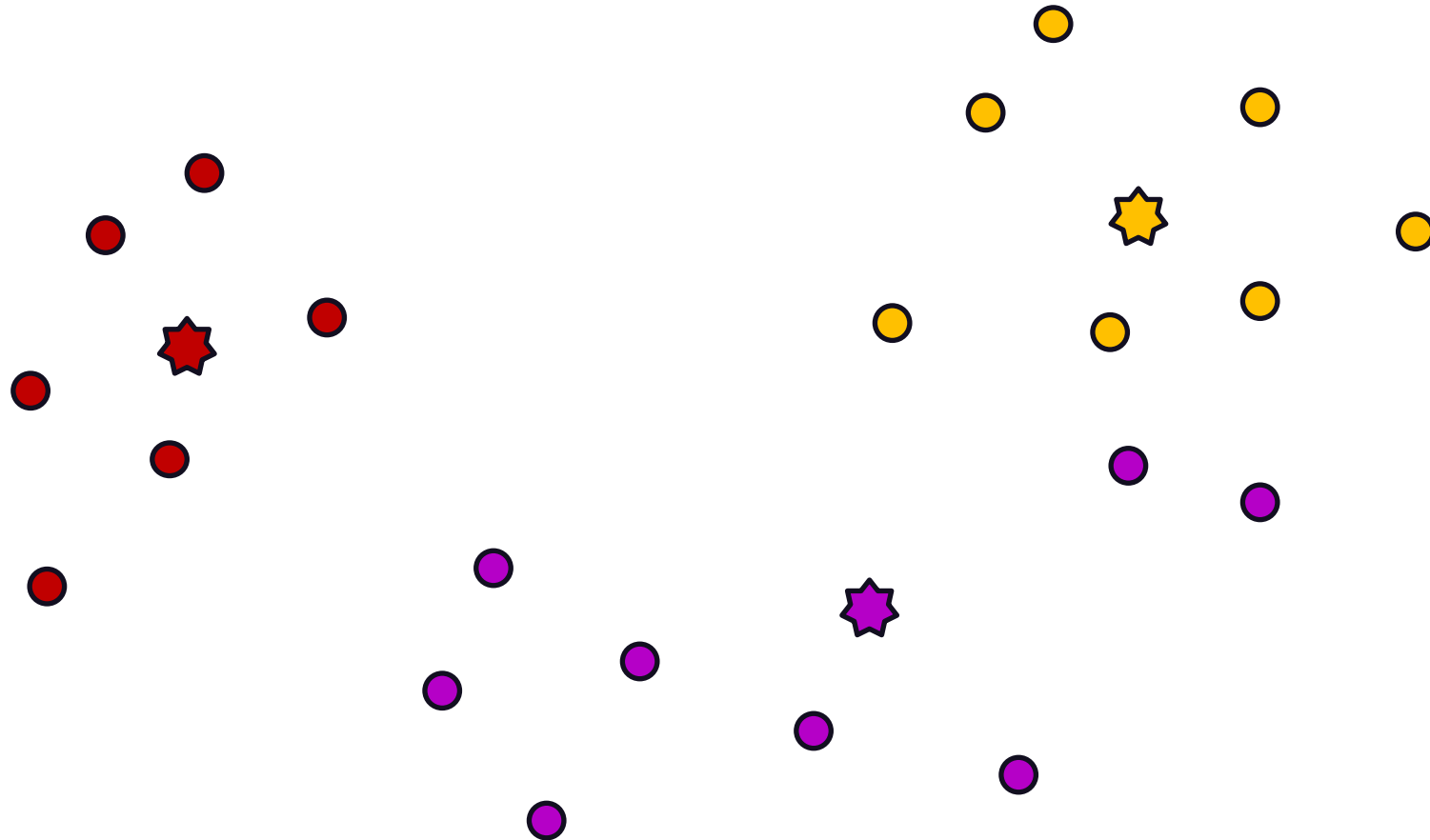
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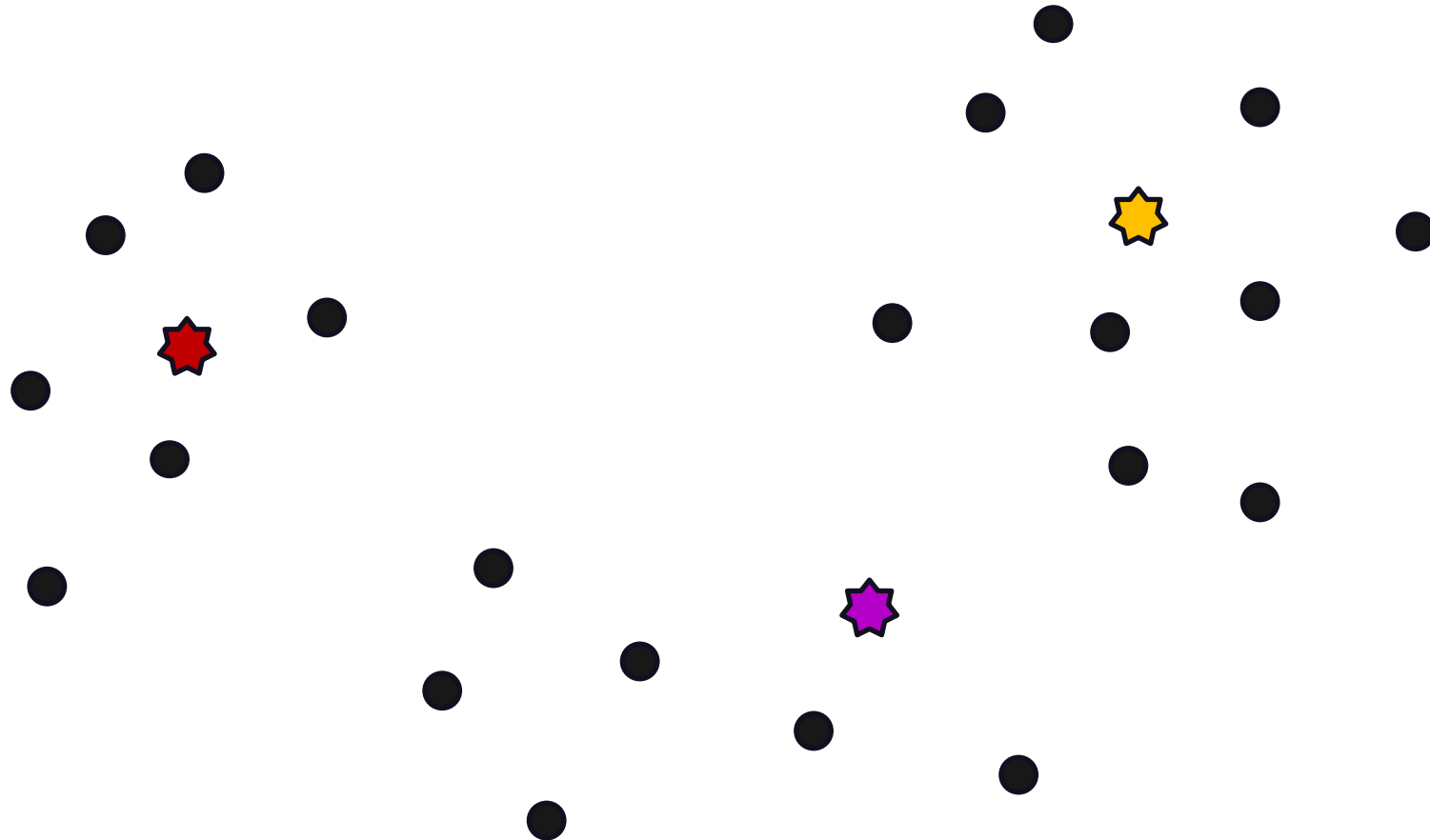
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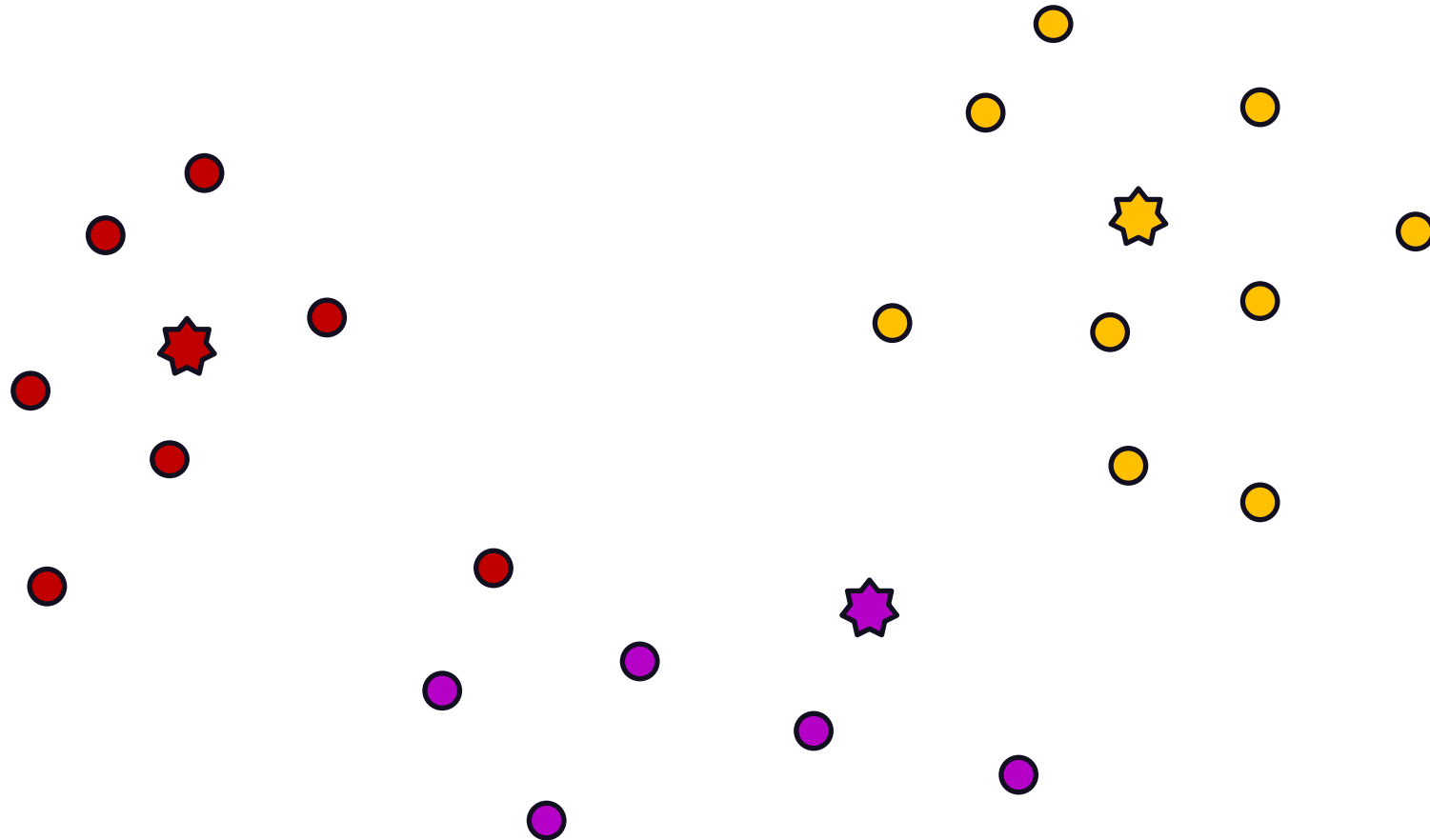
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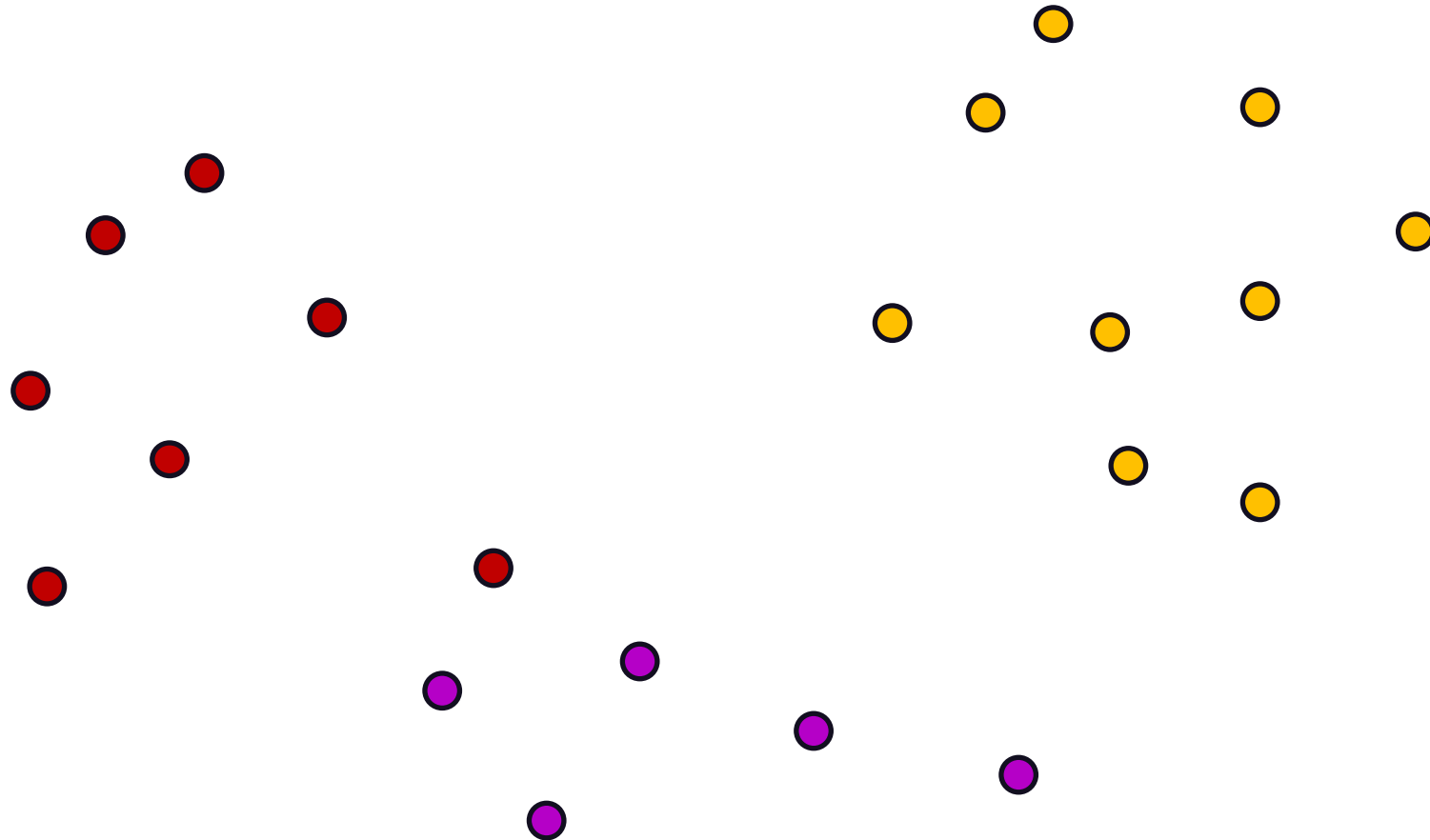
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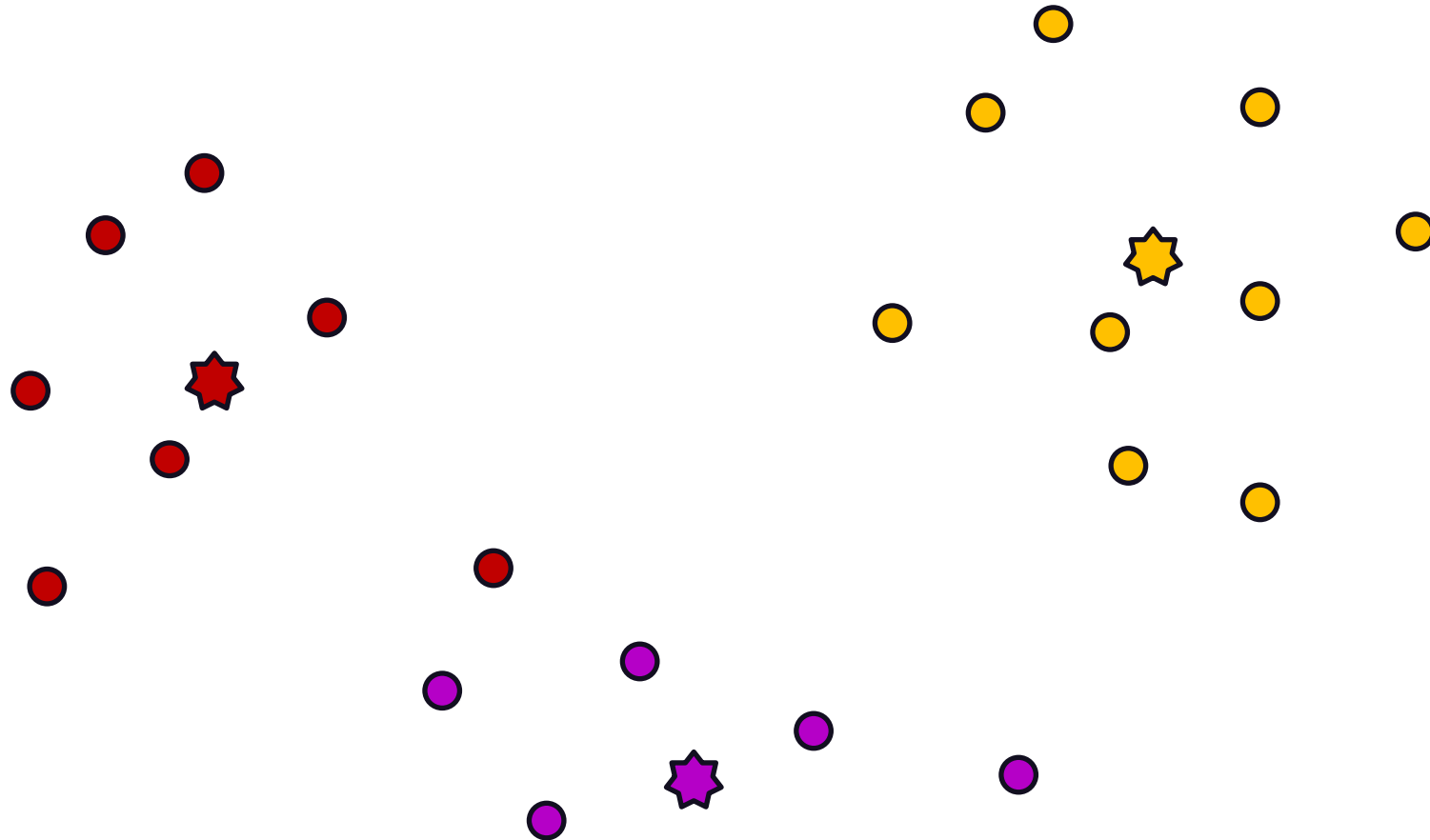
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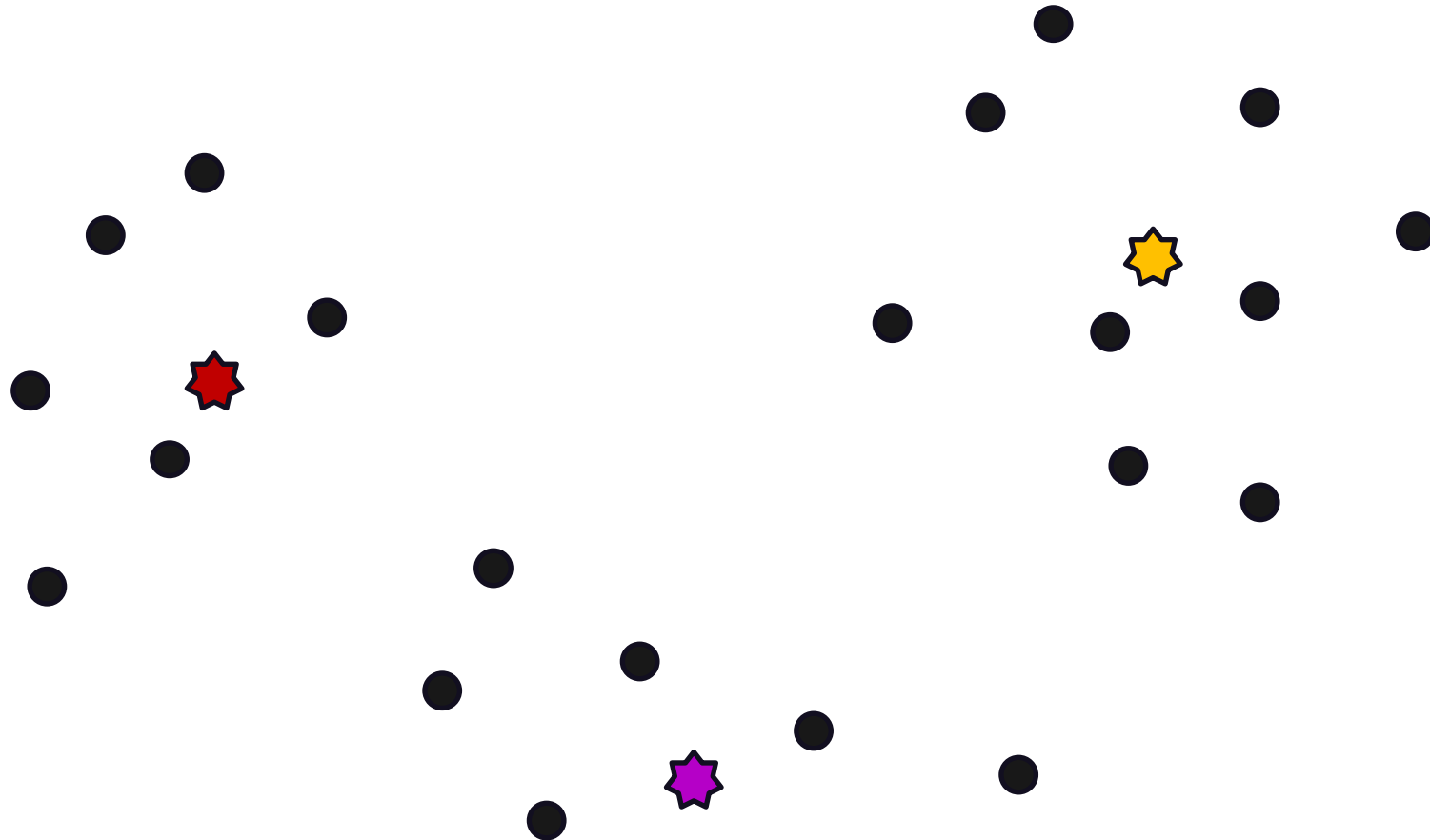
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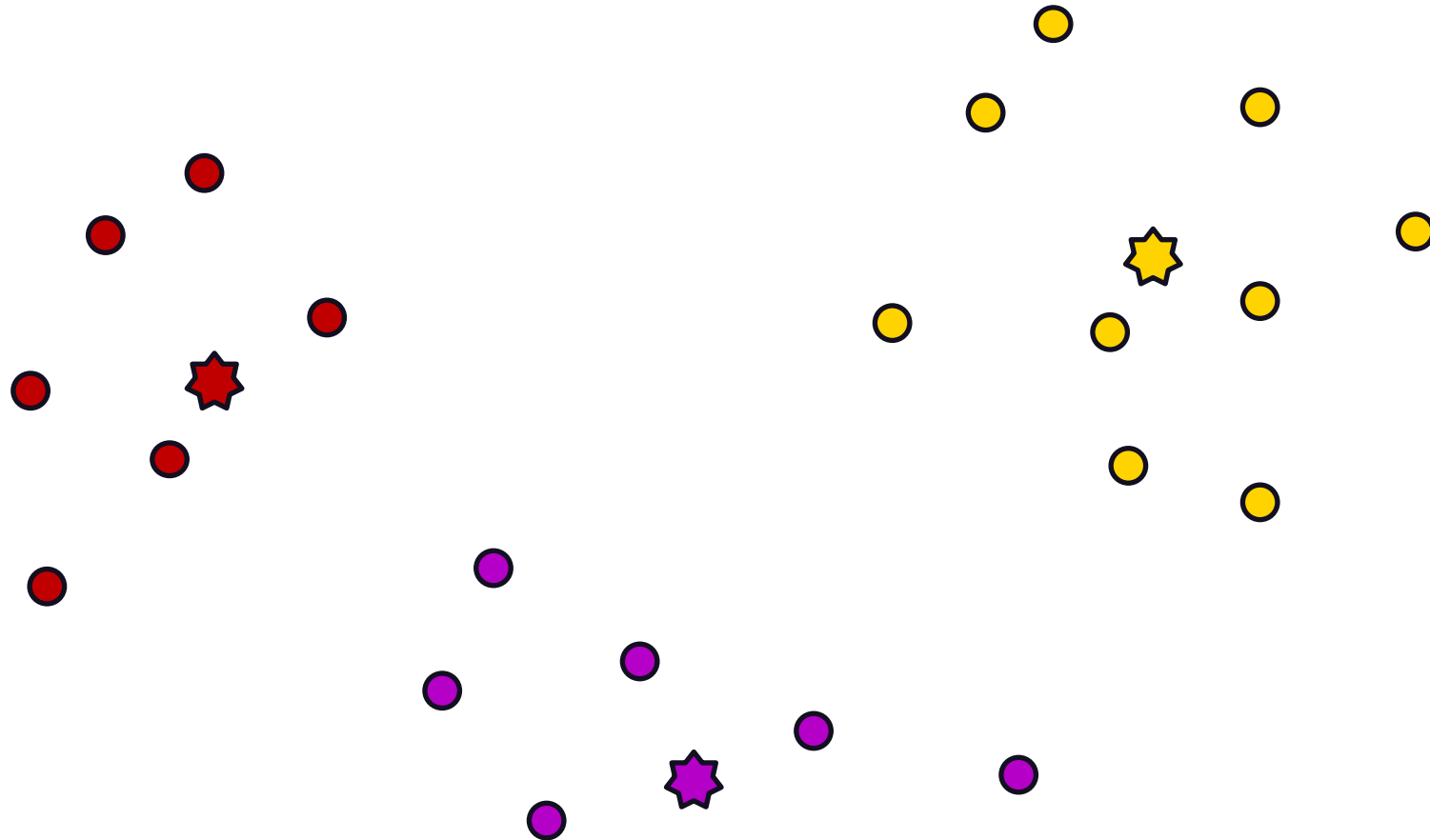
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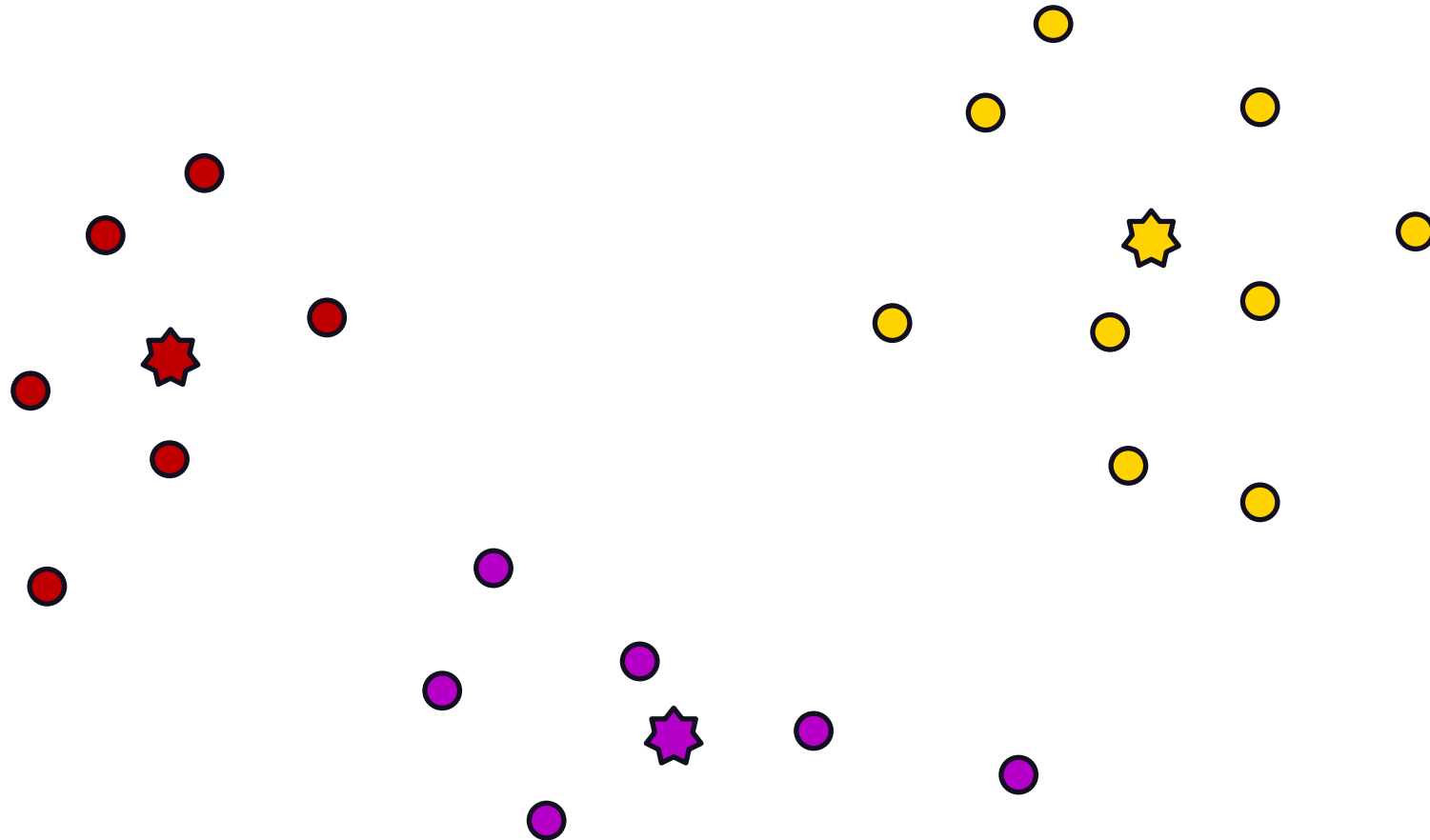
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Representative-based Clustering

The k -Means Algorithm



Representative-based Clustering

The k -Means Algorithm

Advantages

- Conceptually intuitive and relatively **simple to implement**
- k -Means is **scalable to large datasets**.
- Converges to a local optimum, thus **does not produce very poor results**.
- **Domain knowledge can be leveraged** on the initial cluster centers.
- Can **accommodate new observed data** easily.

Representative-based Clustering

The k -Means Algorithm

Disadvantages

- Requires manual specification of number of clusters.
- Result heavily depends on the initial cluster centers.
- Difficult to find clusters of different size, shape, and density.
- Sensitive to outliers
- Noise can dominate on high-dimensional data due to curse of dimensionality

Session 21 and 22 – Representative-based Clustering

Gameplan

Session 21

Introduction to Clustering

Representative-based Clustering

Session 22

Applied Example of Representative-based Clustering

In-Class Activity 10 – Representative-based Clustering

